

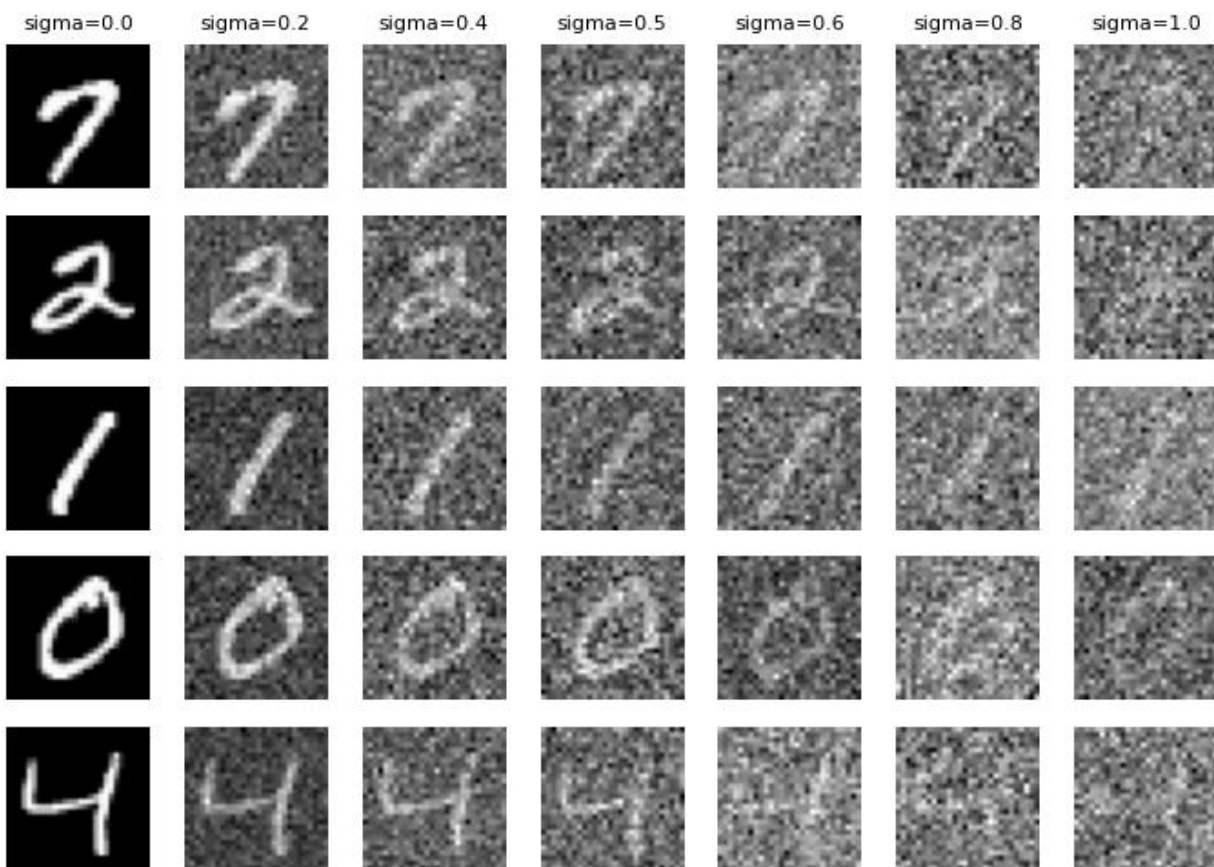
Assignment 5: Diffusion Models

Name(s): Licheng Xu & Libin Wang

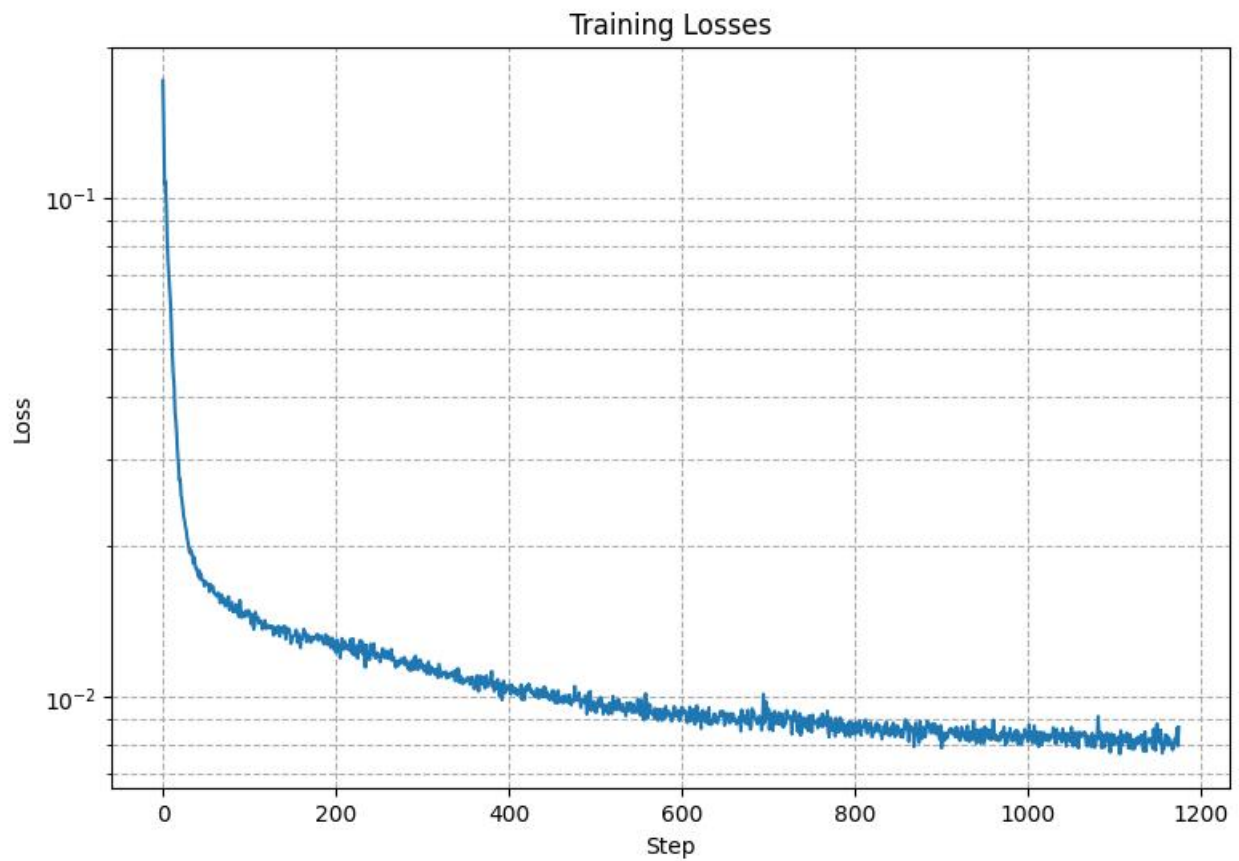
NetID(s): lx8 & libin2

Part 1: Single-Step Denoising UNet

1. A visualization of the noising process using $\sigma = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0]$ (Figure 3)

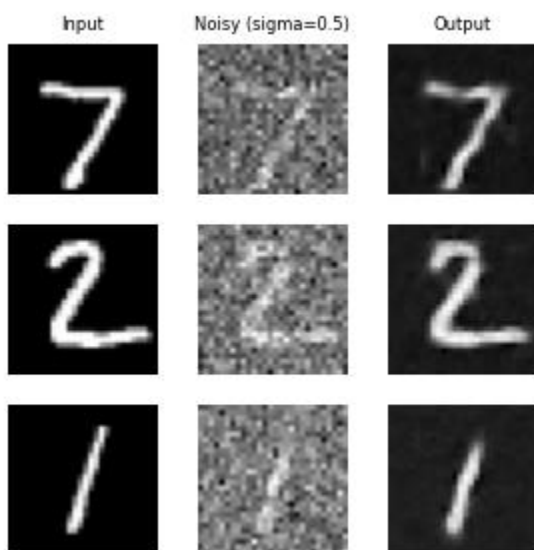


2. A training loss curve plot every few iterations during the whole training process (Figure 4)

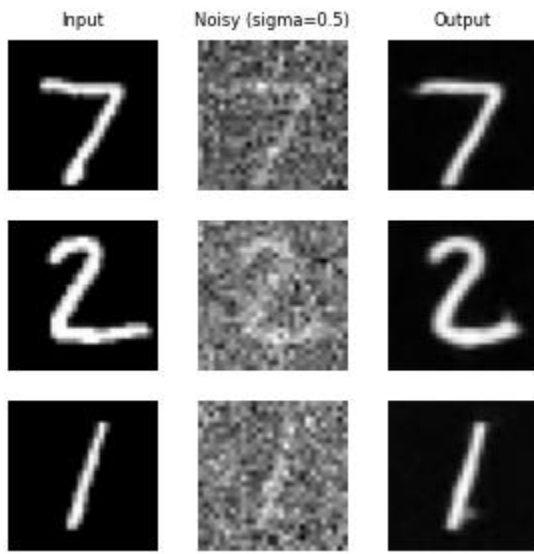


3. Sample results on the test set after the first and the 5th epoch. (Figures 5 and 6)

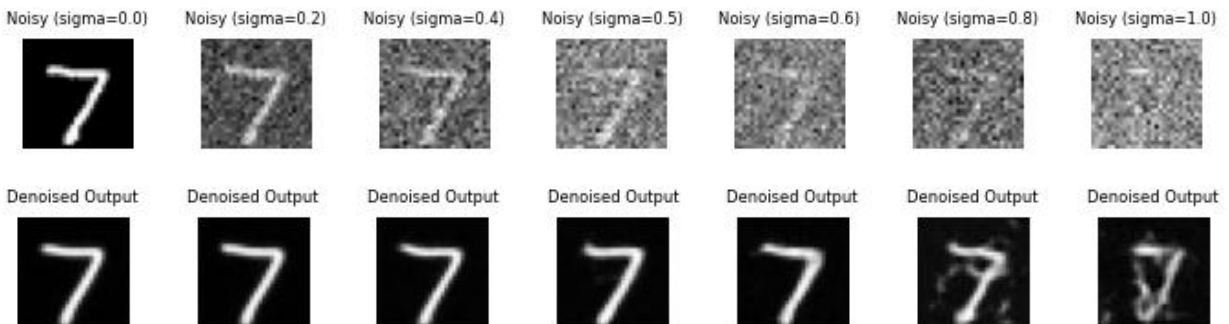
The first epoch:



The 5th epoch:



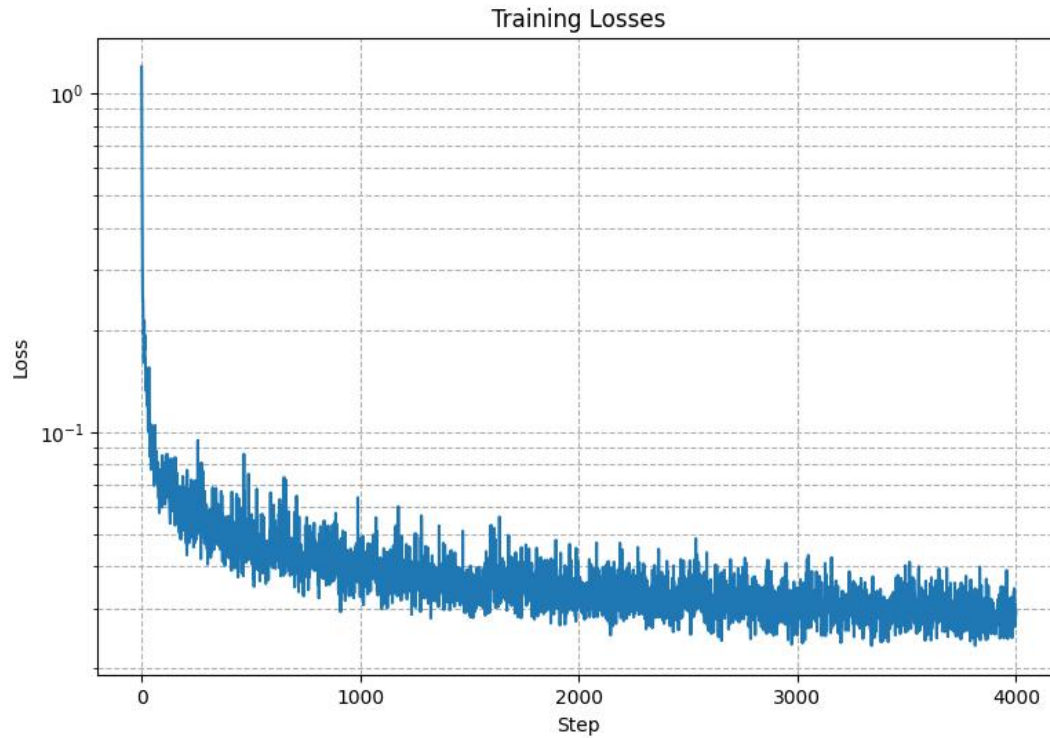
4. Sample results on the test set with out-of-distribution noise levels after the model is trained. Keep the same image and vary $\sigma = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0]$ (Figure 7)



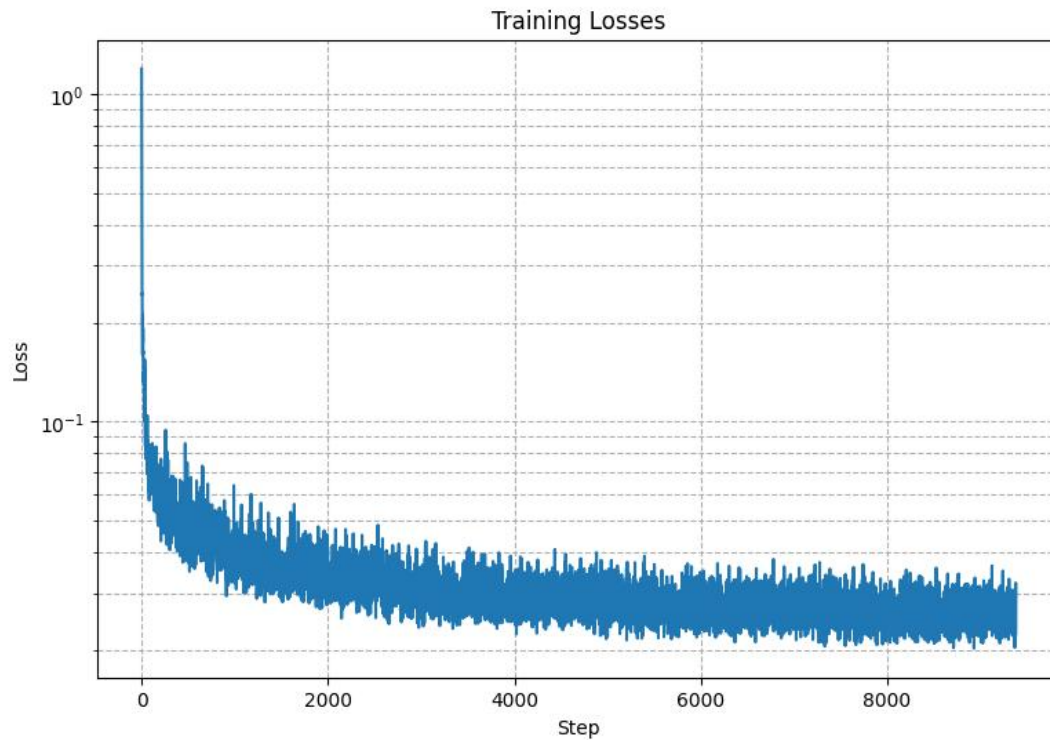
Part 2.1-2.3: Time-conditioned UNet

1. A training loss curve plot for the time-conditioned UNet over the whole training process (Figure 10).

The training loss curve for first 4000 steps:



The training loss curve for whole training process:



2. Sampling results for the class-conditioned UNet after the 5th and the 20th epoch. Providing a GIF is optional and can be done as extra credit.

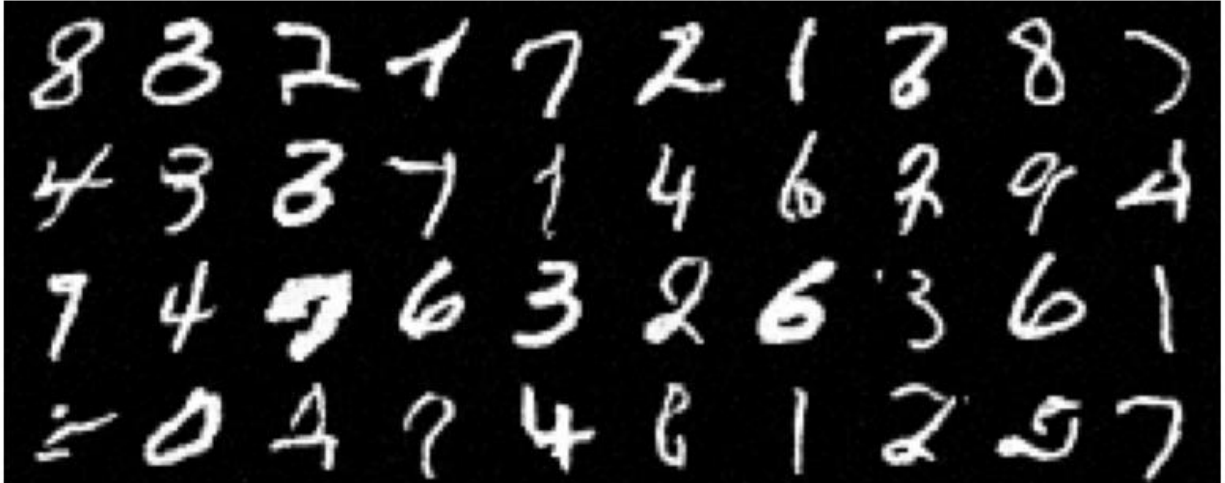
The 5th epoch:

Samples at Epoch 5



The 20th epoch:

Samples at Epoch 20

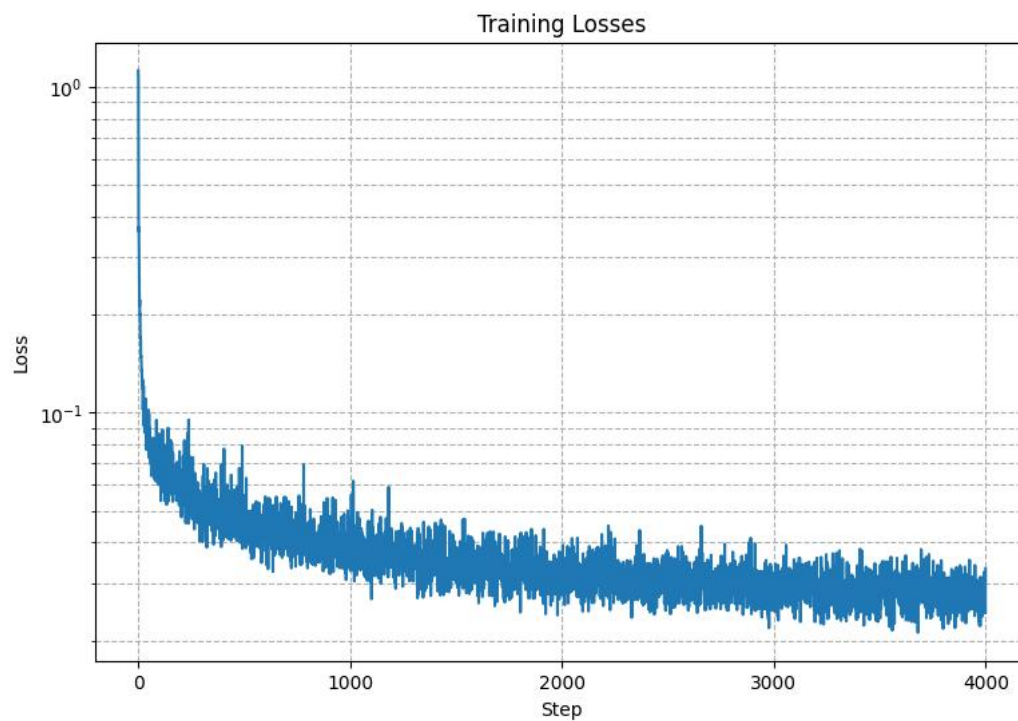


(We generate GIFs for 1st, 5th, 10th, 15th and 20th epoch, included in the .zip file)

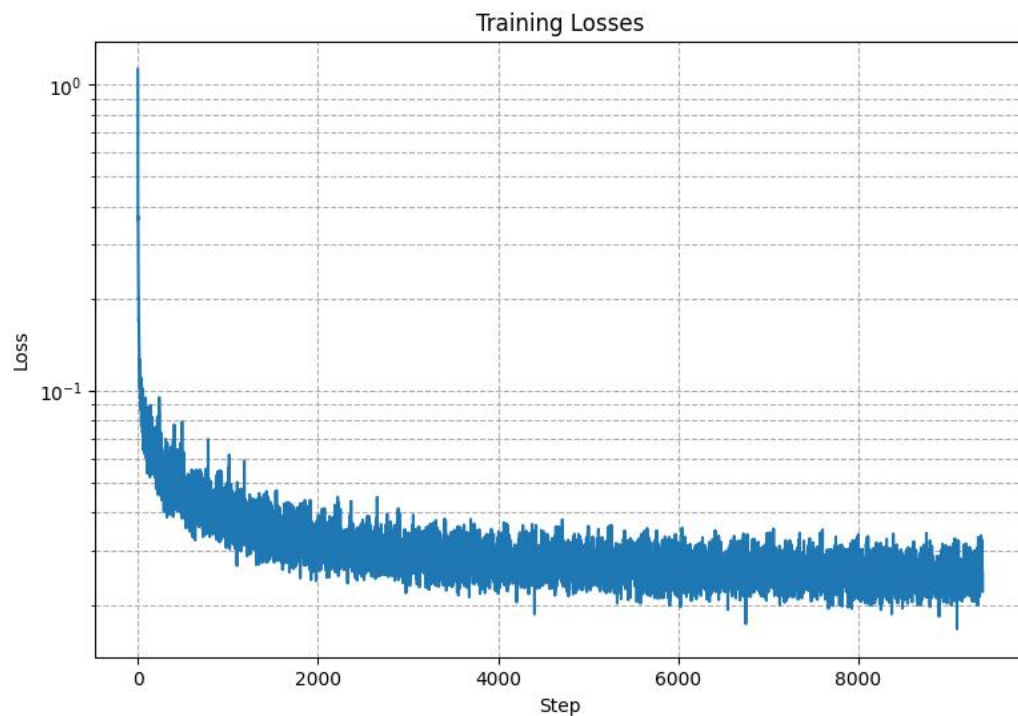
Part 2.4-2.5: Class-conditioned UNet

1. A training loss curve plot for the class-conditioned UNet over the whole training process.

The training loss curve for first 4000 steps:



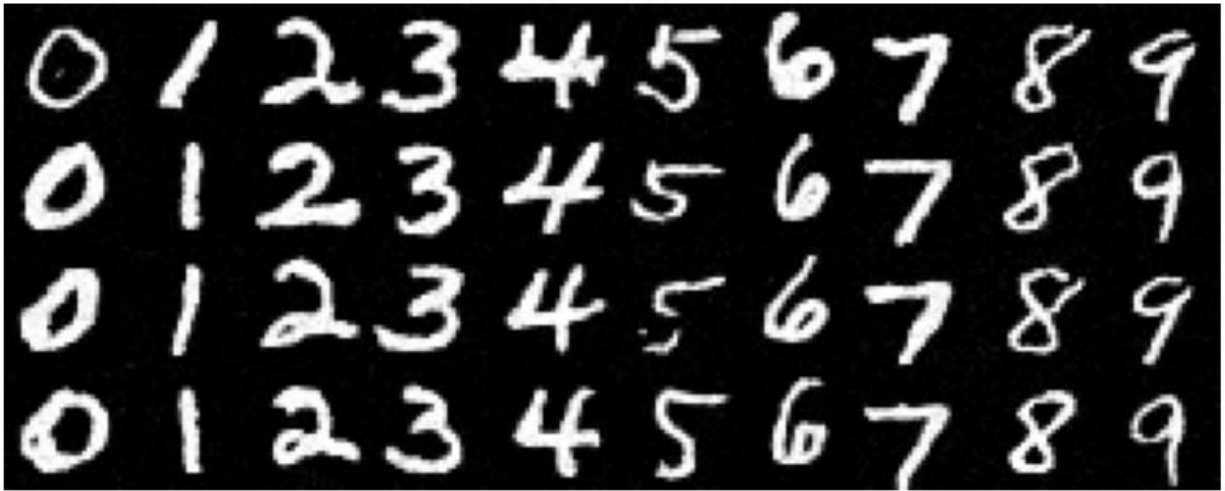
The training loss curve for whole training process:



2. **Sampling results for the class-conditioned UNet after the 5th and the 20th epoch. Generate 4 instances of each digit as shown on the assignment page. Providing a GIF is optional and can be done as extra credit.**

The 5th epoch:

Samples at Epoch 5



The 20th epoch:

Samples at Epoch 20



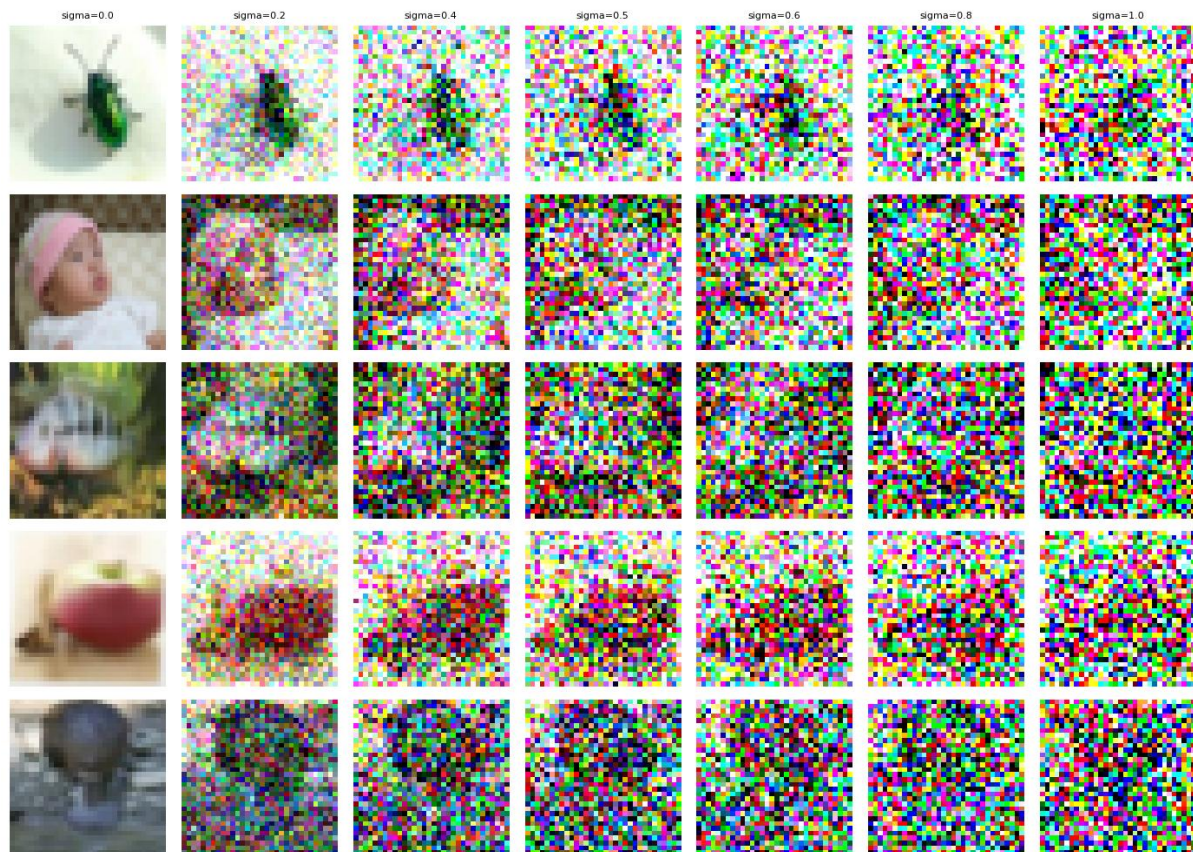
(We generate GIFs for 1st, 5th, 10th, 15th and 20th epoch, included in the .zip file)

Extra credit:

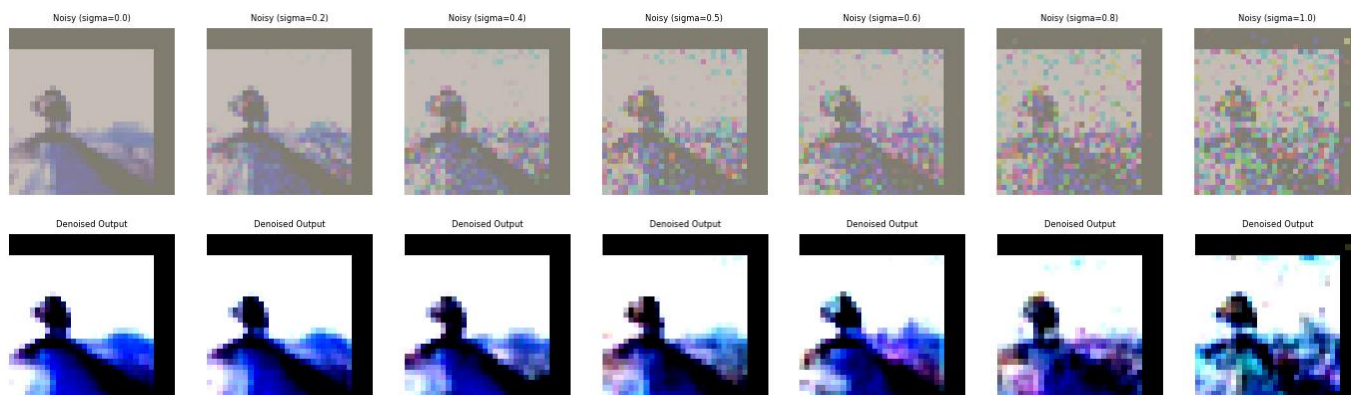
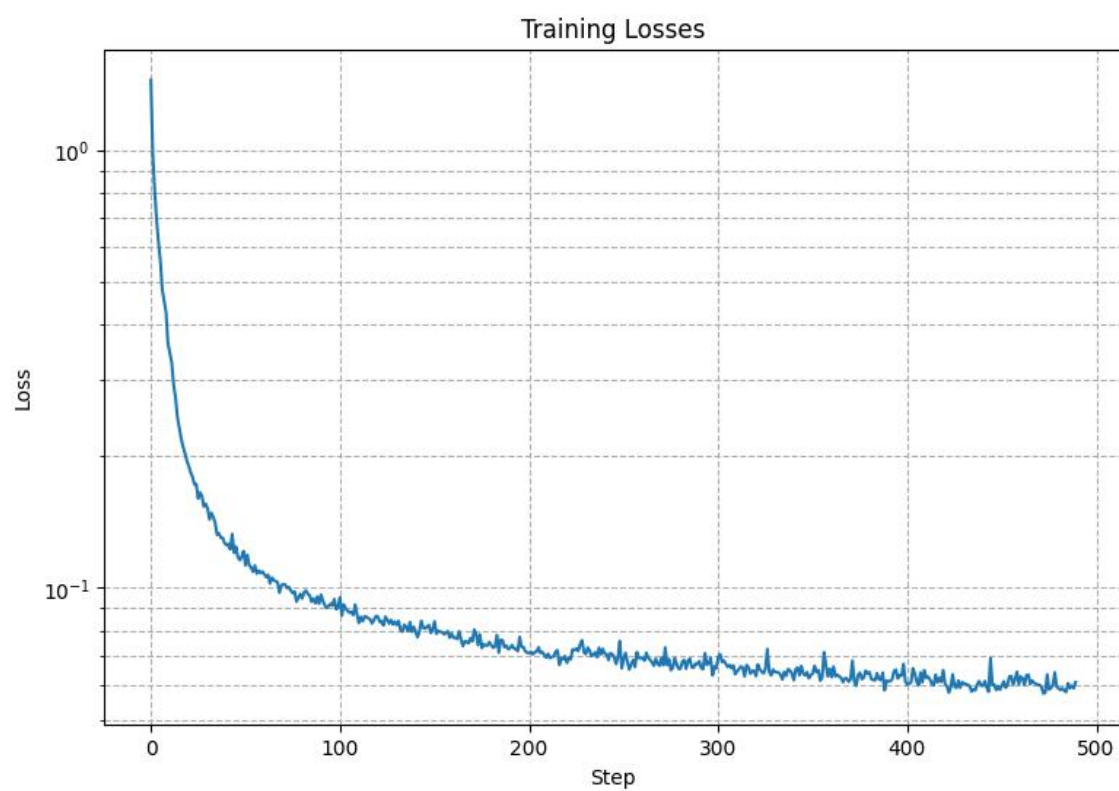
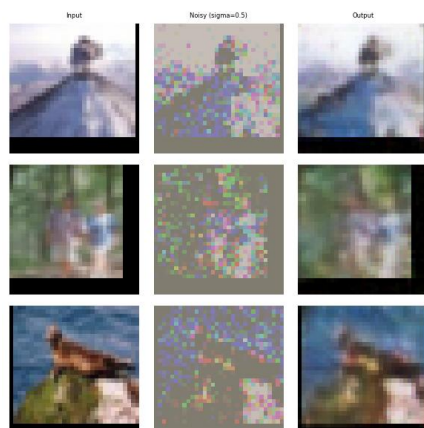
1. We show the process of DDPM and Class-conditioned DDPM, from pure noise to the generated digits images, in .gif format. We choose one time step out of every 10 as a frame, and are generating the corresponding gif images for 1st, 5th, 10th, 15th and 20th epoch. The images are included in the .zip file.

2. We also run the DDPM on CIFAR100 dataset and following is the result.

Same as the MNIST, we also first training a single-step unconditional Unet
The output of the CIFAR100 extra can be seen in the mp5_extra_cifar.ipynb

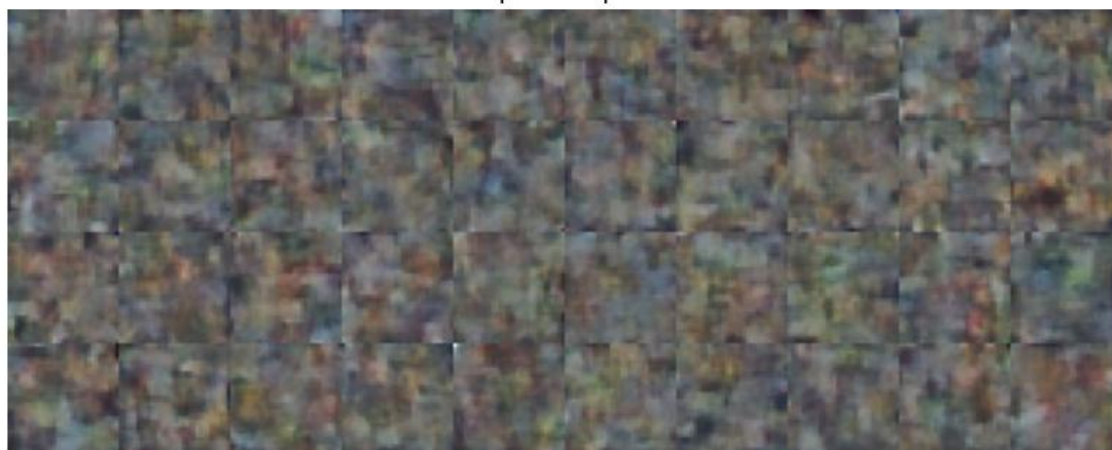
Noisy Sample

Here are the result of the single-step unconditional Unet



Here is the result of the time-conditioned Unet

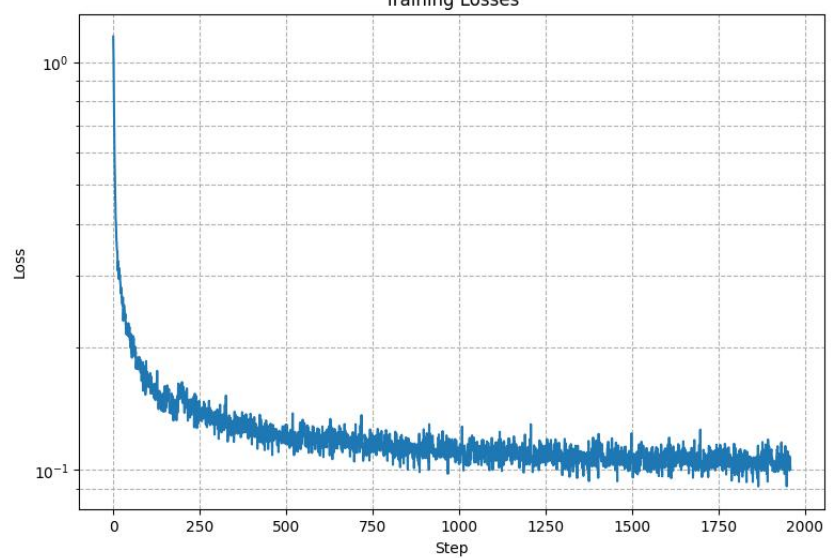
Samples at Epoch 1



Samples at Epoch 20



Training Losses



Here is the result of the class-conditioned Unet

Samples at Epoch 1



Samples at Epoch 20

